

ASHUTOSH MAHAJAN

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[LinkedIn](#) | [GitHub](#) | [HackerRank](#) | [Website](#)



PROFILE SUMMARY

Master’s student in Computer Science at Julius-Maximilians-Universität Würzburg, focusing on Artificial Intelligence and Deep Learning. I bring over four years of software development experience and a strong background in building and improving multilingual and multimodal AI models. My work spans LLM evaluation, generative AI pipelines, and cross-modal learning, using tools like Python, PyTorch, TensorFlow, and Hugging Face. I enjoy connecting solid engineering practices with innovative AI research to create reliable and practical solutions. Deeply interested in applying AI to areas such as medical imaging and intelligent systems, I’m motivated by the challenge of turning complex ideas into impactful, real-world applications.

EDUCATION

Julius Maximilians University of Würzburg Master of Science in Computer Science (Informatik) Grade: 2.0 (in German Scale)	Oct 2023 – Sept 2026 Wuerzburg, Germany
Kalinga Institute of Industrial Technology, Deemed University B. Tech in Computer Science and Engineering Grade: 2.1 (in German Scale)	Jul 2015 – Nov 2019 Bhubaneswar, India

WORK EXPERIENCE

Julius Maximilians University of Würzburg Research Assistant	June 2024 – Present Wuerzburg, Germany
- Developed interactive visualization tools for geometric and multimodal research, supporting experiments in XR and AI-based perception. - Investigated multimodal interaction pipelines (e.g., gesture recognition, cross-input systems), focusing on efficiency and human-machine understanding. - Collaborated on interdisciplinary research connecting XR technologies and AI evaluation frameworks.	
Mobile Premier League Software Development Engineer I	Nov 2021 – Sept 2023 Bengaluru, India
- Engineered large-scale distributed systems with a focus on optimization, fault tolerance, and analytical data processing. - Implemented anomaly detection frameworks using behavioural data and ML-based heuristics. - Applied rigorous debugging and modularization practices to improve reliability and scalability. - Re-architected code for better memory efficiency and modularity across multiple product formats.	
Cosi Games Software Engineer	Feb 2021 – Oct 2021 Bengaluru, India
- Reduced system response latency by 45% through backend optimization and server migration. - Refactored legacy systems, reducing technical debt by 30% while improving long term system stability.	
Zenida Studios Software Developer and Co-Founder	Oct 2018 – Feb 2021 Dehradun, India
- Designed and launched 5+ mobile games featuring intelligent NPC behaviour and adaptive gameplay using heuristic and ML-inspired techniques. - Integrated AI heuristics and third-party SDKs to enhance system functionality and adaptability - Spearheaded cross-functional collaboration to design modular, reusable systems, cutting average development time by 20%.	

PROJECTS

Lexical Meaning Benchmark – Multilingual Semantic QA for LLMs

April 2025 – Aug 2025

Associated with The Julius Maximilians University of Würzburg

[GitHub Repo](#)

- Designed and released a multilingual benchmark covering 50+ languages to assess LLMs’ ability to capture semantic relationships such as hypernymy, meronymy, analogies, and glossary definitions.
- Automated dataset generation by implementing seed-word retrieval and traversal algorithms, integrating BabelNet’s knowledge graph with LM Eval Harness to scale evaluation across five difficulty levels.
- Performed large-scale evaluations and error analyses, examining the impact of language resource availability (high/medium/low), distractor complexity, and LLM characteristics (size, architecture, fine-tuning) on benchmark performance.

Text-to-Image Generation with DeepFloyd IF, Parti, and Imagen

April 2024 – Aug 2024

Associated with The Julius Maximilians University of Würzburg

[GitHub Repo](#)

- Developed a web-based demo for text-to-image generation using DeepFloyd IF, integrated with Hugging Face for accessible deployment.
- Researched and compared state-of-the-art models, including Parti (autoregressive transformers) and Imagen (diffusion-based), with systematic evaluation of DeepFloyd IF for high-fidelity image synthesis.
- Validated model effectiveness by delivering a live demo environment, showcasing real-time text-to-image generation for research and teaching use cases.

Step of Faith: Risk and Fear in Immersive Realities

Oct 2023 – Mar 2024

Associated with The Julius Maximilians University of Würzburg

[GitHub Repo](#)

- Designed and built a VR simulation of a suspended bridge with unstable tiles to investigate fear of heights and falling in immersive environments.
- Analysed psychological and emotional responses to fear-inducing XR conditions, contributing to research on perception and resilience.
- Demonstrated VR’s therapeutic potential by creating controlled exposure therapy scenarios, enabling participants to safely confront phobias and reduce anxiety.
- Explored applications of XR in addressing real-world challenges of perception, safety, and resilience.

SKILLS

Programming Languages: Python, C#, C++ (performance-critical), Java, HTML/CSS

Libraries & Frameworks: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib, Hugging Face

AI/ML Expertise: Artificial Intelligence, Machine Learning, Data Science, AI/ML theories & algorithms, Neural Networks, Model training & evaluation, Deep Learning (CNNs, RNNs, Transformers), Encoder-Decoder Architectures, Large Language Models (LLMs), Vision-Language Models (VLMs), Natural Language Processing (NLP)

Tools & Platforms: Git (Hub, Lab, Enterprise), Docker, Jupyter Notebook, Google Colab, Hugging Face, PyCharm, Unity 3D, Unreal Engine, VS Code, Visual Studio, Windows, macOS, Agile (Jira, Confluence), Microsoft Office, Google Workspace, LM Eval Harness, Postman

Technologies & Workflows: Web Sockets, Object-Oriented Programming (OOP), Modular Software Design, Algorithms & Data Structures, Testing & Debugging, Optimisation, Prompt Engineering, LLM-based Pipelines, API Integrations, Game Development, AI Tools & Workflows, Troubleshooting

Professional Competencies: Leadership, Stakeholder Engagement, Problem-Solving, Critical-Analytical Thinking, Structured Independent Work, Strong Collaboration, Communication Skills (verbal & written), Organisational Skills, Attention to Detail, Adaptability, Fast-Paced Environment Experience

LANGUAGES

English: C1, IELTS 8 BAND

German: Elementary Proficiency, Duolingo Score 40

Hindi: Mother Tongue